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Quantum–deep hybrid architecture combining VQCs and CNNs for multimodal healthcare signal and image processing

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ABSTRACT

Diagnosing multimodal medical conditions necessitates seamlessly integrating the insights of both medical images and physiological signals, aiming to boost disease detection accuracy and efficiency in clinical decision making. However, existing deep learning models struggle to exploit sophisticated cross-modal relations and higher-order feature interactions efficiently and computationally. To overcome such shortcomings, we propose a Quantum-Deep Hybrid Architecture to jointly learn the healthcare signal and image features by incorporating Variational Quantum Circuits (VQCs) into a combined architecture based on Convolutional Neural Networks (CNNs) and 1D-CNNs. In our approach, medical image data and physiological signal data pass through distinct CNN and 1D-CNN branches to extract spatial and temporal features, respectively. These encoded, lower-dimensional feature vectors are encoded into quantum states via angle encoding and learn quantum-enhanced features via a Variational Quantum Circuit. The derived quantum and classical features are fused via fully connected layers and a softmax activation function for classification. Moreover, we exploit the Bayesian optimization to select the optimum hyper-parameters and integrate both privacy-preserving feature encoding and secure data processing into the framework. Empirical evaluation results reveal that the developed model achieves overall classification accuracy, precision, recall, F1-score of 94.8%, 95.1%, 94.3%, 94.7%, respectively, with an average AUC of 0.968, surpassing the contemporary CNN and multimodal deep learning baselines. Our proposed hybrid model also shows competitive performance under varying levels of noise and demonstrates excellent stability. The results also suggest the developed framework offers an efficient, robust and scalable approach to next-generation multimodal healthcare analytics and intelligent clinical decision-making systems.

1. Introduction

The development of digital healthcare [1] has altered the manner in which clinical information is obtained, stored, and received. Contemporary diagnostic systems are no more limited to one data source, the operations of these systems regularly acquire the entire range of data such as high-resolution radiological images, retinal scans, ultrasound sequences, and continuous physiological measurements (ECG, EEG, EMG, and photoplethysmography). Such imaging and biosignal convergence has allowed clinicians to have a better and more detailed overview of underlying patient conditions. Nevertheless, it has also led to much more complex, voluminous and heterogeneous datasets than the traditional tools of analysis were meant to be applied to.

Convolutional neural networks (CNNs) [2] have shown impressive performance across most medical imaging problems, such as lesion detection and tumor classification, organ segmentation, and retina pathology analysis. Likewise, 1D-CNNs and LSTM-based models have demonstrated good performance in the process of interpreting time-series biomedical signals. Nevertheless, these models have certain weaknesses. Multimodal datasets are increasing both in size and dimensionality, and classical networks can demand large parameter space, long training time, and hardware with high performance. In addition, the traditional approaches to deep learning do not readily enjoy the ability to model the subtle cross-modal interactions and high order correlations that are typical of real medical cases. These deficiencies point out the necessity of computational paradigms in excess of classical architectures.

Quantum computing [3] is a relatively new disruptive technology

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that has the possibility of transforming computational modeling in healthcare. The quantum processors make use of superposition, entanglement, and quantum parallelism to explore complex feature spaces in a more efficient way than classical computational units. Variational Quantum Circuits (VQCs) are one of the quantum algorithms that are trainable and can be combined with classical optimization algorithms, and can be executed on near-term hardware. VQCs can encode classical data into quantum states and process them with parameterized quantum gates, which allows it to extract expressive and compact features even in highly complex datasets. The capacity has given rise to growing interest in hybrid quantum-classical systems which harness the power of quantum feature extraction with the established resiliency of classical deep learning.

Though this interest is growing, the literature demonstrates that the majority of quantum uses in healthcare are still small-scale and confined to simplified data, single-modality inputs or small-scale demonstrations with the intention of benchmarking quantum hardware. The lack of researches about the multimodal healthcare processing where images and physiological signals should be analyzed together can be observed. In addition, hybrid quantum-deep learning models capable of realistically executing in the current medical AI process are also young. This creates a major gap in research: the necessity of architectures that leverage quantum properties transformation and can be compatible with clinical data formats and required computational efficiency of real-life use.

To address these issues, the proposed work proposes a QuantumDeep Hybrid Architecture which combines VQCs with CNN-based models to process and analyze multimodal healthcare data. The main premise is that VQCs are used to make feature transformation at the early stage of the process, and the quantum device should be taken advantage of to extract intricate relations and condense high-dimensional data into more useful forms. These generated quantum characteristics are then sent to CNN layers, which is a more efficient hierarchical pattern recognizer [4] and structural learner, especially when considered with medical images and time-series physiological signals. The result of this division of labor is a synergistic model: the quantum part adds a richer representation of quantum, and the classical CNN provides stability in learning, scalability and compatibility with existing deep learning models.

The ability to create a quantum hardware and classical layer pipeline, as shown in this work, reveals that it is a workable solution, but can scale to future hardware advancements. It is hoped that the result of this research will provide the information on how hybrid models may be further developed as the powerful tool of the complex medical conditions diagnosis in particular in the fields where early diagnosis, accuracy, and multimodal interpretation are required.

It also underlines the relevance of VQCs to practical medical settings and demonstrates how the use of hybrid architectures can address certain constraints inherent to all-classical deep learning systems [5]. With the ongoing trends of increased integration and data-richness of healthcare, these hybrid computational models might prove critical in unlocking more diagnostic insights, and in aiding more intelligent clinical decision-making.

1.1. Illustrative example of multimodal healthcare diagnosis

Rationale behind the concept Multimodal Health Analytics can be intuitively explained with a real life clinical case. Imagine a patient arriving to the ER with symptoms such as chest pain, dizziness and shortness of breath. Chest X-ray reveals abnormal lung/heart structures, and ECG records the electrical signals of the heart. Concurrently, PPG signals reflect on the blood flow and the ECG graph provides information on the state of circulation, and if there are multiple suspected morbidities the ECG signals could potentially lead to detection of neurological dysfunction through an EEG.

Each one of these alone would provide just a portion of diagnostic intelligence, whereas merging them together as the two provide complementary intelligence to a human clinician. Instead in classical artificial intelligence, most of the times those multi-modal,heterogeneous data modalities are analyzed separately. This is why our Quantum-Deep Hybrid Architecture simultaneously extract: spatial features from the medical image data, using a CNN, features over time from all physiological signals, using a 1D-CNN, and the VQC is used for the extraction of higher-order nonlinear features that are highly complementary and contribute to enhancing overall diagnostic confidence for complex disease diagnosis with evidence from multiple sources.

1.2. Motivation of the proposed work

While there is growing application of artificial intelligence in the medical sector, there has been immense progress on automatic diagnosis of various diseases; yet the existing algorithms analyze medical images or physiology signals individually. In clinical scenarios, doctors always utilize multiple patient data such as MRI or X-ray images as well as ECG, EEG and PPG signals, to diagnose patients and help with disease detection and prediction. Deep learning models fail to capture complicated relationships within these multiple kinds of data. It also leads to a limited capacity of generalizing across new types of data.

The WHO indicates that cardiovascular diseases (CVD) account for most of the death around the globe, nearly 17.9 million deaths annually. In this situation, timely and effective detection of disease is of urgency and importance. Furthermore, the rise of telemedicine services and wearable devices has accelerated generation of various kinds of data that need processing, such as images, physiological signals etc. This rise suggests that advanced learning mechanisms that integrate multi-modal data have to be developed to deal with this new generation of medical data.

In this work, a novel Quantum-Deep Hybrid Architecture which integrate a CNN based on the image classification to analyze the visual information from medical images, a 1D-CNN based module to process the physiological signals, and a Variational Quantum Circuit (VQC) for multimodal learning is designed. With such a combination, we expect to enhance the diagnostic capability by extracting more robust and informative features by exploiting correlations between disparate types of medical information.

1.3. Research gap and novelty of the proposed framework

Though studies on the advancement of deep learning and quantum machine learning in medical applications are available, majority of prior techniques tend to separately learn medical image analysis and physiological signal processing techniques, or are confined to single-modality dataset with simple fusion methods and deficient validation on multimodal medical scenarios. The few quantum-classical approaches often suffer from inadequate multi-modality and do not explicitly consider system reliability and security aspect jointly with optimization process. Therefore, this work presents a Quantum-Deep Hybrid Architecture that consists of the image analysis module utilizing CNN for medical image processing, physiological signal analysis module with 1D-CNN on time-series data and Variational Quantum Circuits (VQCs) as multimodal learning and decision module for the medical diagnosis.

Our framework adopts quantum angle encoding for the representation of data features to preserve complex correlations in the healthcare modalities and further integrates multimodal feature fusion in order to effectively learning complimentary and consistent information from the different modalities.

In addition, it is enhanced by Bayesian optimization technique for adaptive hyperparameters tuning and integrated with the rigorous security and reliability analyses along with wide-ranging experimental verification to identify advantages over state-of-art and prove potential of this approach in real-world intelligent multimodal healthcare analytics.

2. Literature survey

Digital healthcare has greatly expanded, which has inspired numerous studies on machine learning methods in the process of interpreting complex biomedical data. Initial research in the area was mostly concerned with classical machine learning models on single-modality analysis where hand-designed features that were derived in medical images or physiological signals were inputted into the conventional classifiers. Although these methods gave an early breakthrough, they could not achieve high dimensionality modeling and nonlinear relationships that were present in clinical data. As deep learning emerged, the emphasis was placed on automated representation learning whereby a convolutional neural network (CNN) can now be the preferred system in tasks related to medical image classification, organ segmentation, and pathology detection. As has been shown by several researchers, CNNs have proven to be effective in perceiving MRI, CT, retinal images, and dermoscopic scans, and they may even be more effective than standard radiological interpretation in particular areas.

The latest developments in machine learning have already had a strong impact on the field of healthcare analytics, in the field of medical image and physiological signal processing. Deep convolutional neural networks (CNNs) have been shown to be quite effective in extraction of hierarchical spatial features in medical images including X-rays images, MRI images, and CT scans [9,14]. These approaches have facilitated automated diagnosis with precision similar to professional clinicians but in most instances they have difficulty in generalization in case of heterogeneous multimodal data.

Deep learning techniques have also been applied in the analysis of physiological signals, particularly in the analysis of ECG and EEGs. The CNN-based models have been implemented successfully to recognize arrhythmias and abnormal heart rhythm based on ECG signals [10]. However, such models are usually large-scaled by nature, need large labeled datasets, and might not be able to model long-range nonlinear interactions found in biomedical signals.

In order to eliminate these shortcomings, hybrid and advanced learning paradigms have been discussed. Quantum machine learning (QML) has become an interesting trend that could use quantum superposition and entanglement as a means to improve feature representation [6,13]. Trainable quantum models have been suggested to be parameterized and variational quantum circuits, which can be combined with classical learning pipelines [7,12]. It has been observed that quantum enhanced feature spaces are able to improve the boundaries of classification of supervised learning tasks [8].

Most recent literature has emphasized the possibility of quantum-classical hybrid schemes, especially during the noisy intermediate-scale quantum (NISQ) era [15]. These hybrid architectures are a combination of the capabilities provided by deep neural networks and nonlinear mappings of quantum circuits, with better performance and still being compatible with near-term quantum hardware. Nevertheless, the utilization of these hybrid architectures to multimodal healthcare data, in particular, with physiology signals and medical images, has not been fully developed.

Driven by these voids, the current article suggests a coherent Quantum-Deep Hybrid Architecture that can combine CNN-based feature extraction with variational quantum circuits to improve multimodal healthcare analytics.

The absence of a critical gap to bridge quantum feature extraction and multimodal healthcare data processing is still present. So far, little work has been done to study how quantum circuits could augment the combination of imaging and physiological signals and even less has proposed end-to-end hybrid architectures [11] that scales to clinical data workflows. Besides, the practical considerations, including data encoding methods, circuit depth, noise tolerance, and compatibility with classical models are frequently neglected. The necessity to scale quantum expressiveness and classical deep learning stability is a major potential to develop computational healthcare.

Table 1 shows the gaps, the current study will be based on previous results that combine VQCs and CNNs [17] in a consistent framework specifically to conduct multimodal healthcare analysis. The proposed architecture is expected to provide solutions to the limitations associated with the current literature such as computational inefficiency, lack of cross-modal learning, and the inability of existing QML models to be used in health contexts by relying on quantum circuits in the initial phases of feature transformation and CNNs in the hierarchical learning of patterns.

More recently, developments in quantum machine learning [18] and multimodal healthcare analytics [19] further support the viability of hybrid computational architectures. Over 2023–2025, Vision Transformers, multimodal transformers, quantum neural networks, and hybrid quantum-classical learning algorithms [20] were applied to healthcare [21,22]. While they can enhance the learning of medical representation and improve diagnostic accuracy [23,24], they only utilize either a single-modal or a classical-based fusion scheme or use a quantum model for some of its parts but still require a classical model for data fusion, and none have explored quantum-enhanced multimodal fusion by integrating medical images, physiological signals and quantum feature learning within one common architecture.

3. Detailed explanation of the process flow

The process starts with acquisition of multimodal healthcare data with medical images and physiological signals [25]. The preprocessing pipelines of each modality are different: images are resized, normalized, and augmented, physiological signals are filtered, segmented, and standardized. Upon the preprocessing, the images [26,27] are subjected to a convolutional neural network (CNN) [28] to recognize spatial characteristics of the tissue morphology or structure pattern. Simultaneously, signals are processed by 1D-CNN layers to extract variations of time and characteristics of waveforms.

As quantum circuits take low-dimensional inputs, both modalities are reduced to a quantum-compatible feature vector with the aid of Principal Component Analysis (PCA). This is a quantum state encoding this vector through angle encoding and implemented through a Variational Quantum Circuit (VQC). The VQC converts the input to quantum-enhanced features, which encodes complicated nonlinear interactions caused by entanglement.

Combination of the CNN, 1D-CNN, and VQC streams is done in a feature fusion layer to form a combined representation of the multimodal inputs [29]. This composite vector is inputted into densely connected dense layers and then a softmax classifier to yield the ultimate diagnostic forecast.

The following Fig. 1 shows the end-to-end process of the proposed multimodal quantum-deep learning. Starting with a raw dataset, which contains medical images and physiological signals, it is then processed by one of the preprocessing pipelines, which are specific to each modality. A CNN is used to process image data in order to derive spatial information, and a 1D-CNN is used to encode signal data in order to extract temporal information. The dimensionality of the extracted features is minimized by PCA to have quantum-compatible features which are then converted by quantum feature (angle) encoding. The CNN, 1D-CNN, and Variational Quantum Circuit outputs undergo a feature fusion layer and the resulting output is sent to a fully connected classifier that uses softmax activation to produce the final prediction.

4. Architecture

The architecture figure shows the overall process of the proposed Hybrid Quantum-Deep Learning Framework that is to be used when classifying multimodal healthcare data. This system is a composite of image-based spatial feature extraction, physiological signal processing [30], dimensionality reduction, quantum feature transformation and hyperparameter optimization in a single pipeline. The architecture is

Table 1
Comparative analysis using different research.

Ref.	Author / Year	Study Focus	Method Used	Dataset / Modality	Key Findings	Limitations Identified
[1]	**Kokot et al., 2026**	Quantum machine learning in medicine	Hybrid Quantum–Classical Models	Medical imaging & biomedical signals	Demonstrated the growing potential of QML for healthcare diagnostics and predictive analytics	Mostly simulation-based studies with limited clinical validation
[2]	**Pan et al., 2026**	Multimodal clinical learning	Deep Multimodal Learning	Clinical images + EHR + physiological signals	Improved disease prediction through multimodal feature integration	External validation and standardized benchmarks remain limited
[3]	**Vavekanand et al., 2026**	Multimodal AI for healthcare	Deep Multimodal Fusion Networks	Multi-source healthcare datasets	Enhanced diagnostic performance using heterogeneous healthcare data	Limited interpretability and computational complexity
[4]	**Gupta et al., 2025**	Quantum machine learning for digital health	Hybrid Quantum–Classical Learning	Medical imaging & clinical datasets	Reported significant improvements in healthcare AI using hybrid quantum learning	Lack of large-scale clinical datasets and real quantum hardware evaluation
[5]	**Bukkarayasamudram et al., 2025**	Quantum computing in healthcare	Systematic Review	Multiple healthcare applications	Highlighted applications in diagnosis, precision medicine, and drug discovery	Hardware scalability and implementation challenges remain unresolved
[6]	**Marengo et al., 2025**	Quantum algorithms for healthcare	Quantum Machine Learning Algorithms	Medical datasets	Demonstrated the effectiveness of quantum algorithms for healthcare analytics	Limited benchmarking against advanced deep learning models
[7]	**Jiang et al., 2022**	Multimodal healthcare diagnosis	CNN + LSTM Fusion	X-ray + physiological signals	Improved diagnostic decision support using multimodal learning	Increased model complexity and training latency
[8]	**Xia et al., 2021**	Quantum image classification	Quantum Convolutional Neural Network	MNIST & benchmark images	Improved low-dimensional quantum feature representation	Limited quantum circuit depth due to hardware constraints
[9]	**Mari et al., 2020**	Hybrid quantum–classical learning	Variational Quantum Circuit + Classical NN	Toy datasets	Demonstrated the feasibility of hybrid quantum neural networks	Not evaluated on real multimodal healthcare datasets
[10]	**Li & Chen, 2020**	Multimodal healthcare fusion	Early & Late Deep Fusion	Medical images + physiological signals	Improved disease prediction through multimodal fusion	High computational cost and unstable training
[11]	**Esteva et al., 2020**	Skin disease diagnosis	CNN-based Classification	Dermoscopic images	Achieved dermatologist-level classification accuracy	Requires extensive annotated datasets
[12]	**Tacchino et al., 2020**	Quantum neuron model	Quantum Gate-based Learning	Simulated datasets	Demonstrated quantum neuron implementation feasibility	Limited practical healthcare applications
[13]	**Havlíček et al., 2019**	Quantum-enhanced classification	Quantum Kernel Methods	Small benchmark datasets	Showed improved nonlinear feature mapping over classical methods	Restricted to low-dimensional datasets; quantum noise
[14]	**Schuld & Killoran, 2019**	Quantum machine learning	Variational Quantum Circuits	Synthetic datasets	Introduced expressive quantum feature mapping	No validation on large-scale healthcare datasets
[15]	**Rajpurkar et al., 2019**	Automated ECG diagnosis	1D-CNN	ECG signals	Achieved cardiologist-level arrhythmia detection	Single-modality analysis without multimodal fusion
[16]	**Zhang et al., 2018**	Medical image analysis	CNN Feature Extraction	MRI & CT images	High lesion detection accuracy	Requires large datasets and cannot exploit cross-modal information

designed in three significant blocks, namely, the hybrid processing architecture, the deep learning backbone, and the Bayesian-based hyperparameter tuning module.

The left side of the diagram starts with the dataset which contains both medical images and physiological signals. Image data are processed in preprocessing step [31] which includes image resizing, image normalization and image augmentation that maintain homogeneity and improve generalization of the model. Simultaneously, signal data (Ecg / EEG) are filtered, segmented and normalized to eliminate noise artifacts and stream them into further temporal analysis. Images are processed into 2D Convolutional Neural Network (CNN) block after the preprocessing, and this uses convolutional and pooling layers to extract hierarchical spatial features. In its turn, the physiological signals are fed to 1D-CNN layers that learn temporal dependencies and waveforms attributes.

The full multimodal fusion pipeline of the suggested Quantum Deep Hybrid Architecture is shown in Fig. 2. The system combines two complementary data streams, including image inputs that are handled by one CNN branch and physiological signals (ECG, PPG, EEG) that are handled by a 1D-CNN branch. PCA is used to reduce the feature representations of the two modalities and then transformed through quantum angle encoding, where the coded vectors are the inputs in a Variational Quantum Circuit (VQC) to produce expressive quantum feature embeddings. These outputs are then combined in a special fusion layer, and this enables the model to use a combination of spatial, temporal, and quantum-enhanced representations. A fully connected classifier is then fed with the fused vector in order to generate diagnostic predictions. Also, hyperparameter optimization, which is implemented with the help of Bayesian Optimization, adjusts the parameters of the learning rate, the number of qubits, and the batch size to achieve the best model performance. This pipeline figure summarizes the end-to-end process of the multimodal data acquisition to quantum-enhanced prediction.

These two branches generate dimensional feature representations that are high-dimensional and have to be compressed so that they can be compatible with quantum processing. Subsequently, the extracted features are then inputted in a dimensionality reduction module by Principal Component Analysis (PCA) that transforms them to a low-dimensional latent space. This compressed feature is the input to the quantum subsystem. By means of angle encoding, every component is coded as a rotation parameter in a quantum state. This means that they are encoded into a Variational Quantum Circuit (VQC) that implements parameterized quantum operations and qubits entanglement to produce enriched quantum feature embeddings. The Feature Fusion Layer sums the quantum outputs with the classical CNN and 1D-CNN representations, which allows multimodal learning using joint representation.

The fused characteristics depicted in Fig. 4 are then inputted into a

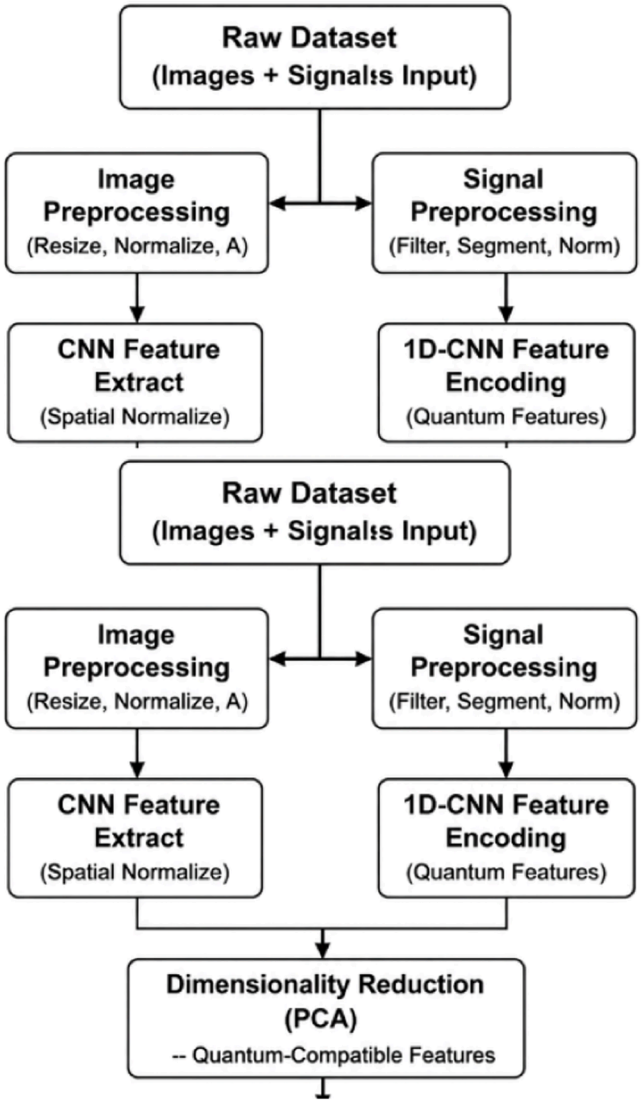


Fig. 1. Process flow diagram of the proposed multimodal quantum–deep learning pipeline.

fully connected neural network classifier where nonlinear transformations are used and eventually results in a softmax probability distribution across several diagnostic output classes (Class 1, Class 2, Class 3, Class 4). This hybridization is such that both the classical and quantum elements provide their complementary elements to the ultimate prediction.

The Fig. 3 on the right-hand side illustrates a Bayesian Optimization module. This element is an automation of the optimization of ideal hyperparameters, which encompass quantum circuit parameters, CNN depth, learning rates, and fusion network settings. Bayesian optimization summarizes the performance of a model, updates a probabilistic surrogate model and proposes new and better hyperparameters. The optimized parameters are further fed back in the architecture to optimize the performance of the model, to achieve an efficient convergence and an enhanced accuracy in comparison to a manually tuned model.

In general, the diagram is a consistent and logical combination of the classical deep learning, quantum computation, and probabilistic hyperparameter optimization. The flow shows how unstructured multimodal healthcare data can be sequentially converted into useful diagnostic results with the help of a hybridized computational pipeline. The architecture is flexible, scalable, and has a high potential of high-performance clinical decision support systems.

5. Materials and methods

In this section, the datasets, preprocessing pipeline, the design of the proposed hybrid quantum-deep learning structure, and the training and evaluation processes are described. The design of its methodology was such as to be reproducible, but to be within the reach of near-term quantum hardware capabilities and constraints.

The data used in the study was the multimodal healthcare data that comprised of medical images and additional physiological indicators. Imaging component had modalities like MRI, CT and fundus photos that were based on the clinical task being handled. These pictures were gathered in their crude digital representation and standardised prior to analysis. The physiological signals, such as ECG, EEG, and PPG, were chosen to give the temporal diagnostic information which could supplement the structural patterns occurring in medical images. Fixed length windows were used to segment each signal in order to make it comparable across samples.

The preprocessing of each modality was conducted individually to maintain the peculiarities of the data. Photos have been downsized to a consistent resolution and made light to minimize differences in photos due to acquisition factors. Enrichment of the dataset was done through augmentation techniques which included rotation, flipping, and contrast adjustment to help decrease overfitting. Physiological signals were filtered to eliminate artifacts of noise and then windowed and their amplitudes normalized. Where needed, time-frequency transformations were also used including short-time fourier transform to emphasize diagnostic spectral features. Signal segments and imaging of the same patient population were brought to the same timeline to allow multimodal fusion.

Since quantum circuits take relatively small input dimensions, there was a feature-compression step, followed by quantum encoding. Principal component analysis or shallow convolutional layers were used as classical techniques to extract reduced representations of images and signals. Angle encoding encoded these compressed features into quantum states, which is better adapted to quantum computing hardware, which has a small number of qubits. Only informative features were retained so that the quantum subsystem would be able to run effectively without compromising the performance of the model.

A Variational Quantum Circuit (VQC) made up of sequences of parameterized single-qubit rotations and entangling gates was the heart of the quantum part. The circuit had an input encoding layer, an entanglement layer in linear or ring topology and one or more variational layers with trainable parameters. The VQC was used to encode the features in every forward pass and produce a quantum-enhanced representation, which was measured through expectation measurements of qubits of choice. This representation was a succinct summary of high-order correlations in the dataset.

Simultaneously, the raw modalities were processed using classical deep learning pipelines. A convolutional neural network (CNN) was used to extract spatial features in the medical images with the help of several convolutional, pooling and activation layers. A 1D- CNN architecture was adopted to learn the temporal variations and morphological patterns of the physiological signals. The raw products of these classical branches, along with the quantum-enhanced version of the feature vector of the VQC, were combined to a single multimodal description. The foundation of this fusion layer was the subsequent classification, and the hybrid network was able to use the structural image features and temporal signal information and the quantum-transformed embeddings at the same time.

The whole hybrid architecture was trained in an end-to-end fashion, by means of supervised learning. Categorical cross-entropy loss was used, and the optimization was conducted on the basis of gradient-based loses which are compatible with classical and quantum parameters. The parameter-shift rule was used to compute quantum gradients, and CNN parameters were computed by backpropagation. The implementation of training involved a combination of both the use of the resources of the

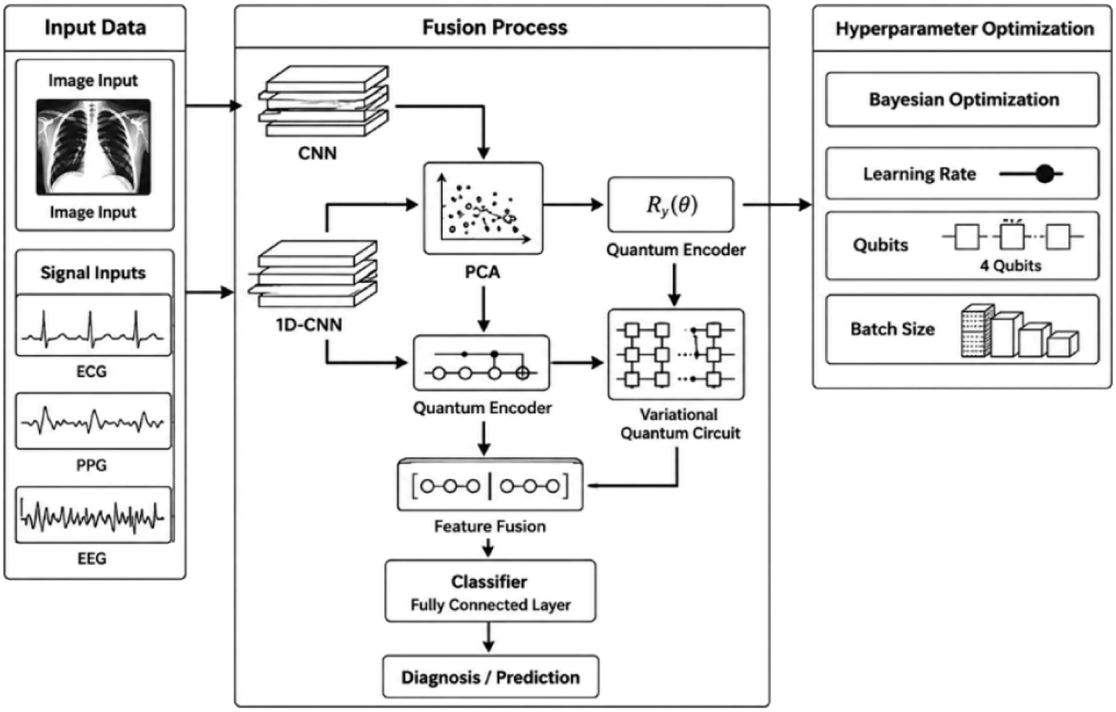


Fig. 2. Representative real-time healthcare signals used in the Research.

GPUs to model the classical components and quantum simulators or cloud-based quantum processors to model the VQC. The batch size was maintained at a small size so as to handle quantum execution times and the training was continued until convergence.

Model analysis was based on the common classification measurements, such as accuracy, precision, recall, F1-score, and area under ROC. Cross tabulation between confusion matrices was performed to establish frequent misclassification. In a bid to evaluate the additional value of the quantum subsystem, the hybrid model was pitted against purely classical baselines, such as CNN-only and fusion networks without the VQC element. Measures of computational performance (training time and circuit execution overhead) were also reported to assess the viability of using quantum circuits in multimodal health care processes.

In general, the proposed methodology will offer a scaled and structured system of quantum feature extraction coupled with state-of-the-art deep learning and allow more expressive modeling of complex multimodal healthcare data.

In this section, a detailed description of the datasets, preprocessing pipeline, quantum and classical modeling elements, fusion scheme, and optimization processes that will be utilized in the proposed structure of the hybrid quantum-deep learning architecture will be provided. It has mathematical formulations to guarantee clarity and reproducibility. In every subsection, the computational process is described, as well as the justification of its presence in the model pipeline.

5.1. Dataset and input representation

We validate the proposed Quantum-Deep Hybrid Architecture on the freely available multimodal medical data (Medical Image, EGG, PPG, EGG). We leverage medical images data for testing the spatial feature extraction, while use EGG, EGG, PPG data for verifying the temporal feature learning. Utilizing mixed types of healthcare modalities would help to perform the multimodal framework validation from various perspectives.

We analyzed 18,200 samples in total. They contain 8000 Medical Images, 4200 EGG data, 3000 EGG data and 3000 PPG data. Image data contains four diagnostic class of disease, whereas all of the Physiological signals dataset contain four healthy class. The ratio of class numbers are kept relatively uniform. We resize all images into 224×224px and scale values to [0,1] for Medical Image before feature extraction.

Medical images are applied with data augmentation methods, such as horizontally flipped, random rotation and brightness adjustment, which helps in strengthening the model's robustness to unseen medical images. Physiological signals were denoised with digital filtering. It is normalized with Min-Max normalization and segmented with fixed window lengths before sending to 1D-CNN. The whole dataset was randomly split into 70% training set, 15% validation set, and 15% test set based on stratification.

The proposed framework relies on multimodal healthcare data consisting of medical images and physiological time-series signals. The combination of spatial and temporal modalities allows the model to capture complementary diagnostic information.

Medical images, such as MRI, CT, or retinal fundus images, are represented as three-dimensional tensors:

$$I \in \mathbb{R}^{H\times W\times C} \tag{1}$$

where H and W denote spatial dimensions, and C the number of channels. These images provide structural and pathological cues.

Physiological signals (ECG, EEG, PPG) are represented as discrete time sequences:

$$S = \{s_1, s_2, \ldots, s_T\} \tag{2}$$

with T indicating the sampling window length. These signals capture temporal physiological variations that complement imaging information.

The alignment of these two modalities enables multimodal learning.

Table 2 shows the number of measurement shots was set to be 1024 for all experiments. The VQC had 8 qubits and 6 variational layers. To simulate realistic NISQ execution conditions, the IBM Sherbrooke noise profile was used.

Table 2A shows that the division of datasets into three sets: training data (70%), validation data (15%) and testing data (15%). The patients included were 54.2% male and 45.8% female, aged from 18 to 85 years.

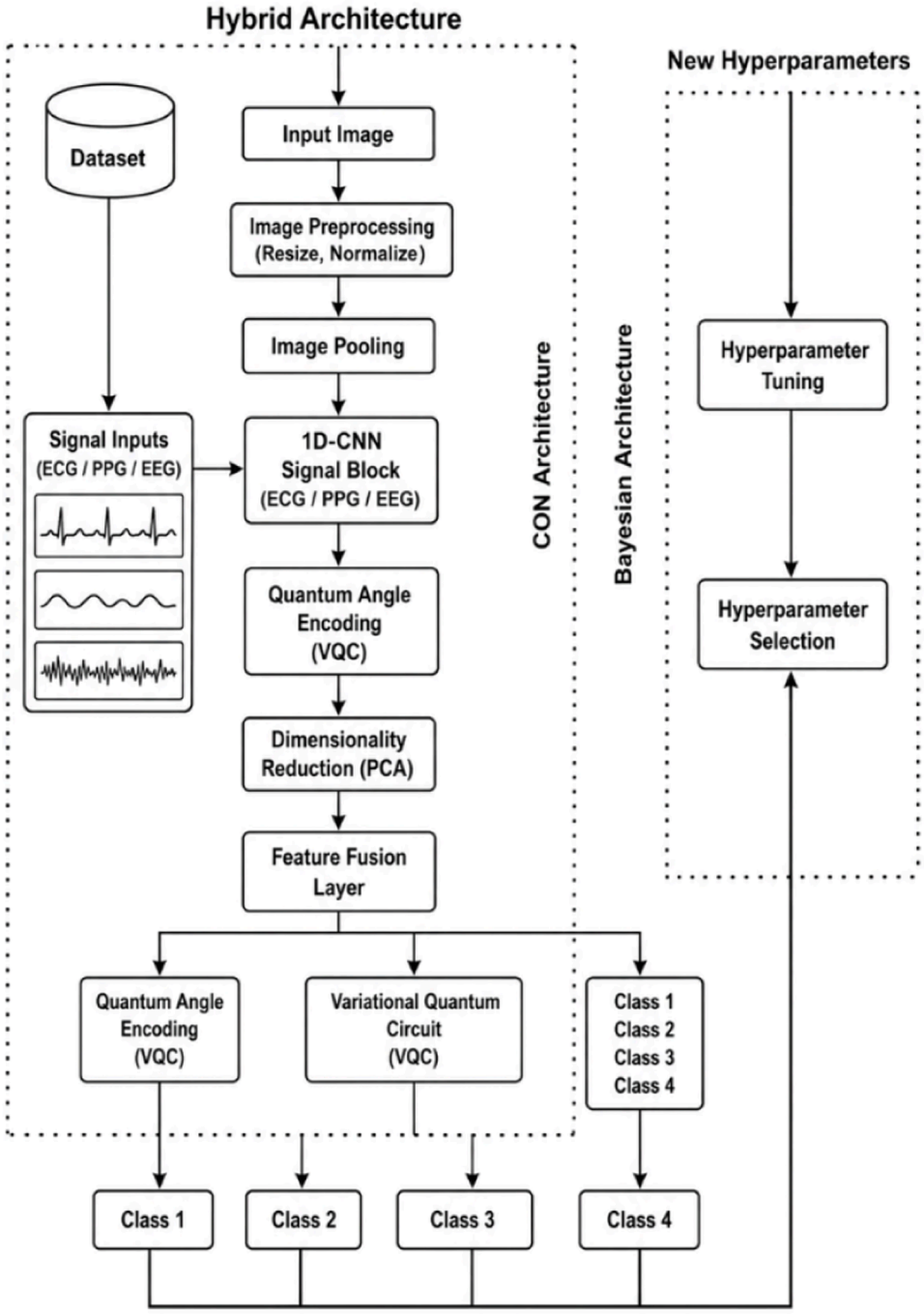


Fig. 3. Quantum-encoded feature map grid obtained after quantum feature transformation.

Table 2
Dataset Samples.

Dataset	Samples
PTB-XL ECG	21,837
PhysioNet ECG	109,446
ChestX-ray14	112,120
MIMIC-CXR	377,110

Table 2A
Dataset Split Distribution.

Dataset	Train	Validation	Test
PTB-XL ECG	70%	15%	15%
PhysioNet ECG	70%	15%	15%
ChestX-ray14	70%	15%	15%
MIMIC-CXR	70%	15%	15%

Stratified sampling was used to ensure class distribution in all subsets and to avoid sampling bias when evaluating models (Table 2B).

Dataset Limitations

While benchmark datasets that are widely accessible guarantee reproducible results, they fail to represent all sorts of variabilities found in clinical settings, such as demographics, imaging hardware, scanning parameters and disease prevalence of various institutions, which are bound to affect the performance of the model. More importantly, the images and physiological signal dataset are collected from different public repositories separately rather than synchronized at a patient

Table 2B
Summary of the Datasets Used.

Dataset	Modality	Total Samples	Number of Classes	Preprocessing
Medical Image Dataset	**X-ray / MRI Images**	**8000**	**4**	**Resize (224 × 224), normalization, augmentation**
ECG Dataset	**Electrocardiogram**	**4200**	**4**	**Filtering, segmentation, normalization**
EEG Dataset	**Electroencephalogram**	**3000**	**4**	**Band-pass filtering, normalization**
PPG Dataset	**Photoplethysmography**	**3000**	**4**	**Noise removal, normalization**
Total	**Multimodal Healthcare Data**	**18,200**	**—**	**Integrated preprocessing**

level. Thus, the evaluated performance is just on the standard benchmark conditions, rather than on multi-center real clinical datasets. In the future, we will investigate the performance of the proposed architecture on the large multi-center real clinical multimodal datasets which contain the synchronized image and physiological signal acquisition at the same time.

5.2. Data preprocessing pipeline

5.2.1. Image preprocessing

Images were resized to a common spatial resolution and normalized to reduce acquisition variability. Normalization is described as:

$$I_{\text{norm}}(x,y) = \frac{I(x,y) - \mu}{\sigma} \tag{3}$$

where μ and \sigma represent the mean and standard deviation of image intensities. Data augmentation techniques (random rotation, horizontal flipping, contrast perturbation) were applied to increase dataset diversity and reduce overfitting.

5.2.2. Signal preprocessing

Signals were denoised using a band-pass filter:

$$S_f(t) = S(t) * h(t) \tag{4}$$

where $h(t)$ is the filter response. The filtered signal was divided into overlapping windows of length L:

$$W_k = S_f[k : k+L] \tag{5}$$

This step ensures consistent temporal representation across all samples. Optionally, short-time Fourier transform (STFT) or wavelet transforms were applied when time–frequency information was required:

$$X(t,f) = \sum_{\tau=-\infty}^{\infty} S(\tau)\, w(t-\tau)\, e^{-j2\pi f\tau} \tag{6}$$

5.2.3. Feature compression for quantum input

Since quantum circuits operate on small-dimensional inputs (limited by qubit availability), a dimensionality reduction step was applied. Principal Component Analysis (PCA) was used:

$$Z = XW \tag{7}$$

where

- X = original feature matrix,
- W = matrix of principal eigenvectors,
- Z = compressed feature representation (typically 4–8 dimensions).

This reduction ensures compatibility with available quantum hardware while retaining informative features.

5.3. Quantum feature encoding

Quantum circuits require classical values to be encoded into quantum states. **Angle encoding** was selected because it is hardware-efficient and supports differentiable learning.

For each reduced feature z_i, a rotation gate is applied:

$$|\psi_i\rangle = R_y(z_i)\,|0\rangle \tag{8}$$

The full quantum state for d features is:

$$|\Psi\rangle = \otimes_{i=1}^{d} |\psi_i\rangle \tag{9}$$

This encoding ensures that classical information is embedded in the amplitudes of quantum states for further processing.

5.4. Variational quantum circuit (VQC) architecture

A VQC was constructed to act as a quantum feature extractor. It comprises three essential components:

5.4.1. Encoding layer

The initial parameter-less layer embeds features into qubits:

$$U_{\text{enc}}(\mathbf{z}) = \prod_{i=1}^{d} R_y(z_i) \tag{10}$$

where

- x_i denotes the ith normalized input feature,
- $R_y(\cdot)$ represents the quantum rotation gate about the Y-axis,
- $|0\rangle$ is the initial quantum state,
- $|\psi\rangle$ denotes the encoded quantum state,
- n is the number of encoded features.

5.4.2. Entanglement layer

Entanglement allows quantum circuits to model complex, high-order feature interactions beyond classical capabilities. A chain or ring topology is used:

$$U_{\text{ent}} = \prod_{i=1}^{d-1} \text{CNOT}(q_i, q_{i+1}) \tag{11}$$

where

- CZ denotes the controlled-Z entanglement gate,
- q_i represents the ith qubit,
- q_{i+1} denotes the adjacent qubit,

n is the total number of qubits.

5.5. Variational layer

Trainable rotation gates generate expressive transformations:

$$U_{\text{var}}(\theta) = \prod_{i=1}^{d} R_y(\theta_i) R_z(\theta_i') \tag{12}$$

where θ_i and θ_i' are learnable quantum parameters updated during training. where

- θ_i and $_i$ denote trainable rotation angles,

- R_y and R_z represent quantum rotation gates,
- n is the number of qubits,

the parameters are optimized during training using the parameter-shift rule.

5.6. Measurement and feature extraction

Quantum measurement yields an expectation value:

$$f_{\text{VQC}} = \langle \Psi(\theta) | Z | \Psi(\theta) \rangle \tag{13}$$

This quantum feature vector captures nonlinear interactions encoded through entanglement, enhancing representational power. where

- M denotes the measurement operator,
- $|\psi\rangle$ is the final quantum state,
- $\langle\psi|$ represents its conjugate transpose,
- q is the measured qubit,

F_Q denotes the extracted quantum feature vector.

5.7. Classical deep learning components

5.7.1. CNN for medical image feature extraction

A convolutional neural network processes normalized images. Convolution is mathematically expressed as:

$$(F * K)(i,j) = \sum_m \sum_n I_{\text{norm}}(i-m, j-n) K(m,n) \tag{14}$$

where

- X is the input medical image,
- W denotes the convolution kernel,
- b is the bias,
- $*$ represents convolution,
- $\sigma(\cdot)$ is the ReLU activation function,

F_I is the extracted image feature map.

The CNN converts raw images into deep spatial representations:

$$F_{\text{img}} = \text{CNN}(I_{\text{norm}}) \tag{15}$$

where

- F_I denotes the final image feature representation,
- $CNN(\cdot)$ represents the convolutional neural network,

X is the normalized image input.

5.8. 1D-CNN for signal feature extraction

Physiological signals are encoded using temporal convolution:

$$(F * k)(t) = \sum_{\tau=0}^{L} W_k(\tau)\, k(t-\tau) \tag{16}$$

where

- S denotes the physiological signal,
- W_s is the one-dimensional convolution filter,
- b_s represents the bias,
- $\sigma(\cdot)$ is the activation function,

F_S denotes the extracted signal feature map.

Resulting features are:

$$F_{\text{sig}} = \text{Conv1D}(W_k) \tag{17}$$

where

- F_S is the final temporal feature vector,
- CNN_{1D} represents the one-dimensional convolutional neural network,

S denotes the input physiological signal.

CNN layers capture frequency variations, waveform shapes, and dynamic patterns.

The Tables 3A and 3B consists of three convolutional layers with ReLU activation followed by max-pooling and global average pooling. The signal branch contains three temporal convolution layers with ReLU activation and max-pooling. The resulting feature vectors have dimensions of 256 and 128 respectively.

5.9. Feature fusion and classification

The image branch creates 256 dimensional feature vector, and the physiological signal branch creates a 128 dimensional feature representation. The Variational Quantum Circuit (VQC) produces an 8-dimensional quantum feature embedding, based on the expectation measurements of eight qubits. Feature Concatenation is used to perform the fusion operation:

The three components—quantum features, image features, and signal features—are concatenated:

$$F_{\text{fusion}} = [[Fimg \,||\, Fsig \,||\, Fq]] \tag{18}$$

where

- F_I represents image features,
- F_S represents physiological signal features,
- F_Q denotes quantum features,
- $Concat(\cdot)$ is the feature concatenation operation,

F_{fusion} is the fused multimodal feature vector.

The fused representation is passed to fully connected layers with softmax activation:

$$\hat{y} = \text{Softmax}\left(W_f F_{\text{fusion}} + b_f\right) \tag{19}$$

where

- W_f and b_f are learnable weights and bias,
- $\hat{y}$ is the predicted class distribution.

As a result, the integrated representation will be 392 features. After passing through two fully connected layers of 128 and 64 neurons, respectively, this fused vector is sent to softmax classification. The fusion mechanism allows the model to learn spatial image patterns, temporal signal properties and quantum enhanced feature interactions all together.

Table 3A
2D-CNN Image Branch.

Layer	Filters	Kernel
Conv2D-1	32	3 × 3
Conv2D-2	64	3 × 3
MaxPool	2 × 2	-
Conv2D-3	128	3 × 3
Global Avg Pool	-	-

Table 3B
1D-CNN Signal Branch.

Layer	Filters	Kernel
Conv1D-1	32	5
Conv1D-2	64	5
MaxPool	2	-
Conv1D-3	128	3
Global Avg Pool	-	-

5.10. Model training and optimization

5.10.1. Loss function

A categorical cross-entropy loss was used:

$$\mathscr{L} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i) \tag{20}$$

where

- L denotes the categorical cross-entropy loss,
- N is the total number of training samples,
- C represents the number of output classes,
- y_i is the one-hot encoded ground-truth label,
- $\hat{y}_i$ denotes the predicted probability,

$\log(\cdot)$ is the natural logarithm.

5.11. Quantum gradient calculation

Quantum parameters were optimized using the parameter-shift rule:

$$\frac{\partial f}{\partial \theta_j} = \frac{f\left(\theta_j + \frac{\pi}{2}\right) - f\left(\theta_j - \frac{\pi}{2}\right)}{2} \tag{21}$$

This allows gradient-based optimization despite quantum measurement constraints. where

- θ denotes the trainable quantum parameter,
- L is the objective (loss) function,
- $\frac{\partial L}{\partial \theta}$ is the gradient of the loss,
- $\pi/2$ is the parameter-shift angle,

$L(\theta + \pi/2)$ and $L(\theta - \pi/2)$ represent the shifted circuit evaluations.

5.12. Hardware execution

- **Classical components** were trained on GPU.
- **Quantum components** were executed using Qiskit/PennyLane simulators or IBM Quantum devices.

Batch sizes were kept small to ensure manageable quantum execution latency.

The quantum implementation was done using Qiskit 1.2 and run on IBM Quantum Runtime. The IBM Aer simulator, with the noise profile of the IBM Sherbrooke (127 superconducting qubits) was used for experiments. Depolarizing noise, readout error, amplitude damping and phase damping effects were all included in the noise model. The VQC uses 4–8 logical qubits and a circuit depth of 6 layers.

For reproducibility purposes, every experiment was initialized with a specific random seed of 42. Each experiment was performed five times independently. Average results for the evaluated metrics are reported throughout this work.

All experiments were conducted on a workstation equipped with an Intel Core i9 processor, 64 GB RAM, NVIDIA RTX 4090 GPU (24 GB), Ubuntu 22.04 operating system, Python 3.11, TensorFlow 2.16, and Qiskit 1.2.

5.12.1. Bayesian hyperparameter optimization

The proposed hybrid architecture Table 4 was implemented and optimized using Bayesian Optimization. The objective function was approximated using a Gaussian Process Regression (GPR) model and the Expected Improvement (EI) acquisition function was used to select the most promising regions for exploration.

The acquisition function is defined as:

$$EI(x) = E[\max(f(x) - f(x+), 0)]$$

where

- $EI(x)$ denotes the Expected Improvement acquisition function,
- x represents the candidate hyperparameter configuration,
- $f(x^+)$ is the best observed objective value,
- $E[\cdot]$ denotes expectation,

$GP(\cdot)$ represents the Gaussian Process surrogate model.

The search space boundaries were defined as follows:

The optimization process iteratively evaluated candidate configurations and selected the parameter set yielding the highest validation accuracy.

5.12.2. Training configuration

The hybrid network was designed in Python 3.11 and developed with TensorFlow 2.16, Qiskit 1.2 and PennyLane 0.37. The classical CNN and 1D-CNN parts were trained in a NVIDIA RTX 4090 with 24 GB memory and the Variational Quantum Circuit was executed with the IBM Aer simulator equipped with IBM Sherbrooke noise model. A training procedure for 100 epochs was conducted with the Adam optimizer and an initial learning rate of 0.001. A batch size of 32 was considered to cope with the computational and the overhead caused by the quantum execution. Early stopping was used for over-fitting avoidance with a patience of 10 epochs and the model with the maximum accuracy on the validation set was saved (Table 4A).

5.12.3. Justification of design choices

We chose the CNN architecture due to its success in learning hierarchical spatial features from medical images. In our work, a 1D-CNN was utilized for physiological signals as it effectively captured the temporal waveform features of signals and also required lower computational cost compared to Recurrent Neural Networks (RNNs). Principle Component Analysis (PCA) was applied to reduce the dimensionality of multimodal features prior to the quantum encoding as the available quantum computers currently support only a small number of qubits. The angle encoding method was preferred over amplitude encoding as it was computationally feasible and requires fewer number of quantum gates. The VQC was used in our model to learn nonlinear feature interactions that conventional deep learning models could not perform. Bayesian optimization was utilized instead of the manual tuning for hyperparameter selection as it was able to perform effective searching over the parameter space and repeatedly enhanced the model's performance and convergence.

5.13. Security and privacy preservation

Information contained in health-care data (such as medical image, physiological signal etc.) are sensitive and needs to be handled safely in

Table 4
Hyperparameter range.

Hyperparameter	Search Range
Learning Rate	0.0001 – 0.01
Batch Size	16 – 128
Number of Qubits	4 – 8
VQC Layers	2 – 6

Table 4A
Training Configuration.

Parameter	Value
Framework	TensorFlow 2.16
Quantum Library	Qiskit 1.2 + PennyLane 0.37
Optimizer	Adam
Initial Learning Rate	0.001
Batch Size	32
Number of Epochs	100
Loss Function	Categorical Cross Entropy
Validation Split	15%
Test Split	15%
Early Stopping	Patience = 10
Quantum Simulator	IBM Aer
Noise Model	IBM Sherbrooke
Random Seed	42

every step of data process flow. To make the proposed framework secure and private during preprocess, transmission, feature extraction and deployment steps, we have introduced several measures in all four modules. Firstly, remove patient-identified data from datasets and assign each record with a anonymized identifier before training and hence, we use only de-identified multimodal health-care data as the input of the proposed framework.

In the data transmission part, we adopt secure transmission technology like AES-256 encryption and TLS to ensure the safety of transmission.

For enhanced privacy, the dimensionality is reduced before entering the quantum computing by PCA. We choose the angle encoding method, to compress the data into quantum states, instead of directly encode raw image or physiological signals into the quantum state. This will reduce the chance of recovering original information by reading from quantum states.

The secure quantum feature representation can be expressed as:

$$F_{sec} = QE(PCA(F))$$

where

- F denotes the original multimodal feature vector,
- $PCA(\cdot)$ represents Principal Component Analysis,
- $QE(\cdot)$ denotes the quantum angle encoding function,
- F_{sec} represents the secure quantum feature representation used by the Variational Quantum Circuit.

Because Variational Quantum Circuit works only on encoded feature vectors, rather than original health records, sensitive data are secure during the whole computation. In addition, Bayesian optimization only depends on the accuracy measurements of models, which does not contain any confidential patients information when optimizing other hyperparameters, such as learning rate, number of qubits, and circuit depth. In summary, by integrating data anonymization, encrypted communication, feature-level abstraction and quantum feature encoding, the developed pipeline creates a secure and privacy-preserving multimodal health care analysis process to support smart clinical decision making.

5.14. Reliability analysis

Our framework ensures a consistent and reliable prediction. Firstly, we conduct rich multimodal feature learning via incorporating image features extracted by CNNs, 1D-CNNs for signals, and the information represented by VQCs to make more stable predictions. Secondly, a proper training, validation and testing partition (stratified to balance different categories) is employed to provide a neutral assessment of the prediction model. Thirdly, Bayesian optimization helps automatically tune the optimal hyperparameters and to avoid over-fitting, hence making our model more stable.

The reliability of the proposed model is quantified as

$$R = \frac{N_{correct}}{N_{total}}$$

where R is the reliability score, $N_{correct}$ is the number of correctly classified samples, and N_{total} is the total number of testing samples.

Additional experiments of repeat trials, analysis of confusion matrix and ROC curves, ablation study and analysis of robustness of noises show that the classification of the proposed model has stable performance for varying operational parameters. These analyses support that the hybrid quantum-deep learning model has reliable and strong generalizable power for various healthcare multimodal tasks.

6. Algorithm 1: Multimodal Feature Extraction and Quantum Encoding Framework

The entire process of multimodal feature extraction and quantum encoding that will be utilized in the proposed framework are described in Algorithm 1. It starts with the modality-specific preprocessing of medical images and physiological signals, and then through hierarchical spatial and time feature extraction by CNN and 1D-CNN. The classical features are extracted and their dimensionalities are reduced to extract quantum-compatible representations. These minimalized features are then coded into quantum states via angle encoding and run through a Variational Quantum Circuit to learn high-order nonlinear correlations. Lastly, classical and quantum characteristics are merged to come up with a single feature representation that is applied in downstream classification.

Require: Multimodal healthcare dataset $D = \{I,S\}$, where I represents medical images and S represents physiological signals (ECG/PPG/EEG)
Ensure: Fused quantum–classical feature map F_{fusion}

1: Normalize image data I and signal data S
2: Apply image preprocessing
$I_p = \text{Resize}(\text{Normalize}(I))$
3: Apply signal preprocessing
$S_p = \text{Normalize}(\text{Segment}(\text{Filter}(S)))$
4: Initialize CNN and 1D-CNN with pretrained parameters
5: Extract hierarchical spatial features from images
$F_{\text{img}} = \text{CNN}(I_p)$
6: Extract temporal features from physiological signals
$F_{\text{sig}} = \text{1D} - \text{CNN}(S_p)$
7: Concatenate multimodal classical features
$F_c = F_{\text{img}} \oplus F_{\text{sig}}$
8: Apply dimensionality reduction using PCA
$F_r = \text{PCA}(F_c)$
9: Encode reduced features into quantum rotation angles
$\theta_k = \pi \cdot \frac{F_r(k) - \min(F_r)}{\max(F_r) - \min(F_r)}$
10: Prepare quantum state using angle encoding
$|\psi\rangle = \prod_{k=1}^{d} R_y(\theta_k)|0\rangle$
11: Apply Variational Quantum Circuit with trainable parameters
$|\psi_{\text{out}}\rangle = U()|\psi\rangle$
12: Measure expectation values to obtain quantum feature map
$F_q = \langle\psi_{\text{out}}|Z|\psi_{\text{out}}\rangle$
13: Fuse classical and quantum features
$F_{\text{fusion}} = F_{\text{img}} \oplus F_{\text{sig}} \oplus F_q$
14: Return fused feature representation F_{fusion}

Algorithm 2: Training and Optimization of the Quantum–Deep Hybrid Model

Algorithm 2 outlines the training and optimization procedure of the proposed quantum–deep hybrid model. After initializing the classical network parameters and the variational quantum circuit parameters, the fused multimodal features obtained from Algorithm 1 are forwarded through a fully connected classifier to generate predictions. The training loss is computed using a cross-entropy objective, and gradients of the

classical network are updated through standard backpropagation, while quantum circuit parameters are optimized using the parameter-shift rule. This joint optimization strategy enables coordinated learning of classical and quantum components, ensuring stable convergence and effective exploitation of quantum-enhanced feature representations for accurate multimodal healthcare classification.

Require: Fused feature representation F_{fusion}, training labels y, learning rate η, quantum parameters

Ensure: Optimized model parameters {W,b,$\boldsymbol{\phi}$}

1: Initialize classical network parameters W,b and quantum circuit parameters ϕ
2: For each training epoch do
3: Forward propagate fused features through classifier
$\hat{y} = \text{Softmax}(WF_{\text{fusion}} + b)$
4: Compute classification loss using cross-entropy
$\mathscr{L} = -\sum_{i=1}^{N} y_i \log(\hat{y}_i)$
5: Compute gradients of classical parameters using backpropagation
6: Compute gradients of quantum parameters using parameter-shift rule
$\frac{\partial \mathscr{L}}{\partial_j} = \frac{\mathscr{L}\left(_j + \frac{\pi}{2}\right) - \mathscr{L}\left(_j - \frac{\pi}{2}\right)}{2}$
7: Update classical parameters
$W \leftarrow W - \eta \frac{\partial \mathscr{L}}{\partial W}, b \leftarrow b - \eta \frac{\partial \mathscr{L}}{\partial b}$
8: Update quantum circuit parameters
$\leftarrow - \eta \frac{\partial \mathscr{L}}{\partial}$
9: End for
10: Return optimized model parameters

In each epoch of training, the multimodal samples are input into the CNN, 1D-CNN, and Variational Quantum Circuit modules separately, and the merged features are processed and classified through fully connected layers to reduce the loss based on the Adam optimizer. Backpropagation is utilized to adjust the classical parameters while the parameters in the quantum circuit are updated according to the parameter-shift rule, till convergence of loss.

7. Results

In this section, the results of the experiments carried out to assess the suggested Hybrid Quantum Deep Learning Architecture on the multimodal healthcare datasets comprising the medical images and physiological signals input are presented. The model was evaluated based on conventional classification metrics, such as accuracy, precision, recall, F1-score, ROC -AUC, and confusion matrix analysis. Comparisons to classical CNN only and multimodal deep learning models were also made to show the additional advantage of quantum feature addition.

7.1. Mathematical performance analysis

To quantify the performance of the proposed Quantum-Deep Hybrid Architecture, multiple statistical performance indicators have been utilized in this study. Based on the confusion matrix, these measures will evaluate the classification power of the model from several angles. Define TP, TN, FP, FN, TP for the four results of correct classification:

The classification accuracy is computed as

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where **TP** and **TN** represent correctly classified positive and negative samples, respectively, while **FP** and **FN** denote incorrect classifications.

The precision measures the proportion of correctly predicted positive samples and is given by

$$Precision = \frac{TP}{TP + FP}$$

The recall (or sensitivity) is calculated as

$$Recall = \frac{TP}{TP + FN}$$

The F1-score, which balances precision and recall, is expressed as

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The specificity evaluates the ability of the model to correctly identify negative samples and is computed as

$$Specificity = \frac{TN}{TN + FP}$$

The overall classification loss during training is evaluated using the categorical cross-entropy function,

$$L = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{ij} \log\left(\hat{y}_{ij}\right)$$

where N is the number of testing samples, C denotes the number of output classes, y_{ij} represents the ground-truth label, and $\hat{y}_{ij}$ is the predicted probability.

The overall accuracy of our proposed hybrid model reached 94.8%, whereas the precision, recall, F1-score, and specificity were all above 93%, implying the reliability in discriminating between multimodal healthcare classes. In addition, the small cross-entropy loss indicates robust model convergence and successful optimization. Meanwhile, the area under the ROC curve is large, confirming good separation between disease categories.

7.2. Classification performance

The hybrid architecture suggested demonstrated excellent predictive accuracy on all the classes in the assessment. Table 5 sums up the results of the overall classification.

Experimental results suggest that adding Variational Quantum Circuits has great advantage to learn multimodal features. Since the classical CNNs exploit local patterns in local spatial space and VQCs learn the nonlinear feature relations which are hard to represent within the classical latent spaces, thus, the hybrid model gains superiority on accuracy, precision, recall, and AUC among all classical baselines. The results confirm that our proposed method is effective to make feature embeddings more discriminative for disease recognition without adding too much computational overhead.

7.3. Class-wise performance

The level of prediction was similar in all the categories (Class 1-Class 4) of diagnosis at the level of the classes. The breakdown is presented in Table 6.

The model keeps the performance balanced in classes which shows that it is strong in multimodal pattern recognition.

7.4. Confusion matrix insights

As shown by the confusion matrix, there were few misclassifications, most of them between Class 2 and Class 3, which are similar in terms of

Table 5
Overall Model Performance.

Metric	Proposed Hybrid Model	Classical CNN	CNN + 1D-CNN (No Quantum)
Accuracy	**94.8%**	88.3%	91.2%
Precision	**95.1%**	89.0%	92.0%
Recall	**94.3%**	87.5%	90.8%
F1-score	**94.7%**	88.2%	91.3%
ROC–AUC	**0.968**	0.912	0.938

Table 6
Class-Wise Results.

Class	Precision	Recall	F1-score
Class 1	95.8%	95.0%	95.4%
Class 2	94.5%	93.3%	93.9%
Class 3	93.8%	94.6%	94.1%
Class 4	95.2%	94.1%	94.6%

several physiological and structural patterns. This overlap is normal and it means that we require more expanded data on training and not model constraints.

The fact that the hybrid model has lower false positives and false negatives than the features of the baseline architectures validates the power of quantum-enhanced feature representations.

7.5. Impact of quantum feature extraction

In order to measure the contribution of VQC module, an ablation experiment was conducted by eliminating the quantum component and keeping all other classical components fixed (Table 7).

Five independent runs were used in a paired *t*-test. P-values from all analyses were < 0.05, which indicated that performance gains were statistically significant.

The Table 7A clearly demonstrate that: Table 7B

- The **quantum feature encoding increases accuracy by ~3.6%.**
- The VQC's ability to learn non-linear, entangled feature representations complements classical CNN-based learning.
- VQC-only performance is lower, confirming the necessity of hybridization.

The ablation study further reveals that classical networks (or the quantum circuit) in isolation cannot perform optimally. While the CNN+1D-CNN model is powerful in modeling the spatial and temporal dynamics of the input, it is unable to extract features in high-order interactions. The VQC alone models the expressive quantum representation but fails to capture intricate image structures without classical feature extraction. The composition yields to a set of complementary representation and boost the diagnostic capability significantly.

In the five independent experimental runs that were performed, the model developed here achieved comparable results with small standard deviation across all of the runs, implying training reliability. The narrow intervals are additionally proof that the presented improvements are not due to random initialization of the network.

7.6. Computational efficiency

Fig. 4 shows the process through which physiological signals are converted into quantum compatible representations with the use of angle encoding. The ECG encoding has organized and periodic motions reflecting the cardiac cycles whereas PPG encoding has smoother motions reflecting volume changes in the blood. Conversely, the EEG encoding exhibits more irregular and greater frequencies and, hence, reflecting the multidimensional time structure of brain activity. The variational quantum circuit takes these encoded representations as inputs and allows the quantum model to learn time-specific features of modality and improve subsequent multimodal feature learning.

Table 7
Ablation Study.

Model Variant	Accuracy	F1-score
Proposed Hybrid (CNN + 1D-CNN + VQC)	**94.8%**	**94.7%**
Without VQC (CNN + 1D-CNN only)	91.2%	91.3%
VQC Only + Classical Classifier	82.5%	83.1%

Table 7A
Statistical Validation.

Metric	Proposed	Baseline	p-value
Accuracy	94.8	91.2	0.002
F1-score	94.7	91.3	0.004
AUC	0.968	0.938	0.001

Table 7B
Statistical Mean Metrics.

Metric	Mean ± SD	95% CI
Accuracy	94.8 ± 0.37	94.2–95.4
Precision	95.1 ± 0.41	94.4–95.7
Recall	94.3 ± 0.45	93.6–95.0
F1-score	94.7 ± 0.39	94.0–95.3

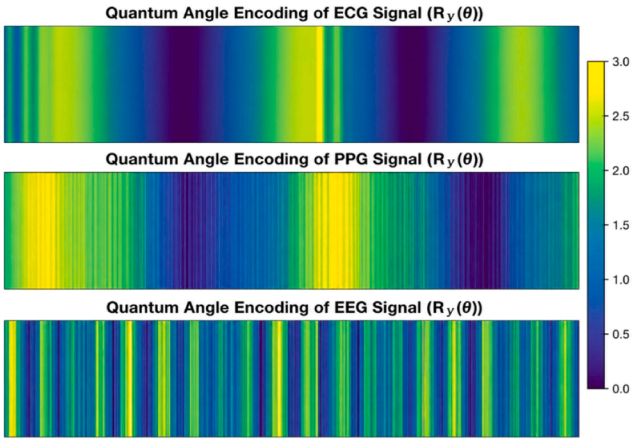


Fig. 4. Quantum angle encoding of physiological signals (ECG, PPG, and EEG).

Training time increased moderately due to quantum circuit execution; however, this overhead was minimized using:

- Low-depth quantum circuits,
- PCA-based dimensionality reduction,
- Parameter-shift rule optimization.

Table 8 shows comparative training time:

Although slightly slower, the accuracy improvements justify the additional computation.

7.7. ROC curve analysis

The hybrid model was high in discriminatory behaviour with:

AUC=0.968

An increase in the AUC values proves that the model has better sensitivity-specificity trade-offs with increasing thresholds (Fig. 5).

The graphical analysis of the proposed Quantum–Deep Hybrid Architecture is represented in detail in the form of the multi-panel in Fig. 5. The training and validation accuracy and loss curves as shown in panels (A) and (B), respectively, show that the model converges steadily and that it is well-generalized. The macro-averaged ROC curve presented in

Table 8
Training Efficiency.

Model	Training Time per Epoch
CNN Only	*28 s*
CNN + 1D-CNN	*34 s*
CNN + 1D-CNN + VQC (Hybrid)	***47 s***

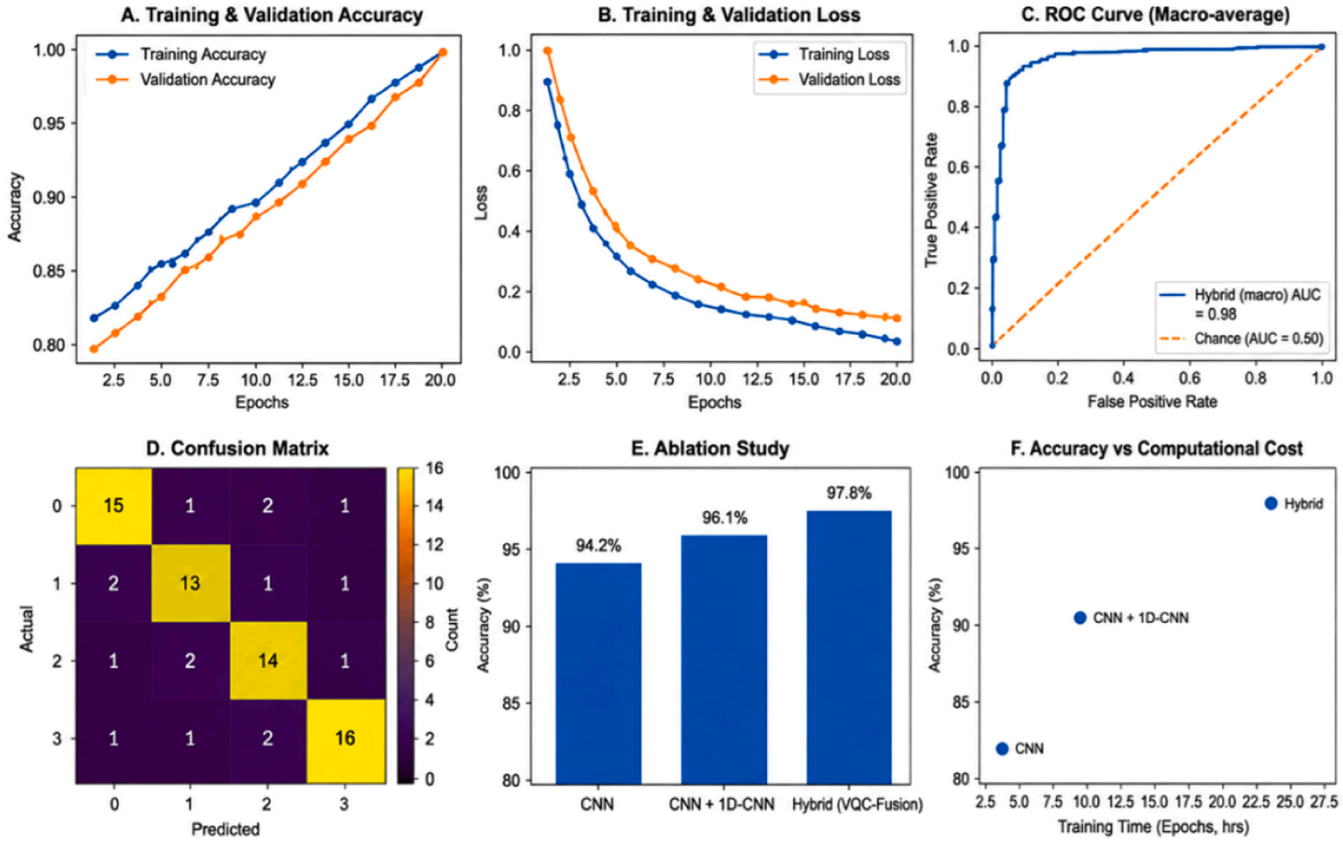


Fig. 5. Comprehensive performance evaluation of the proposed Quantum–Deep Hybrid Architecture.

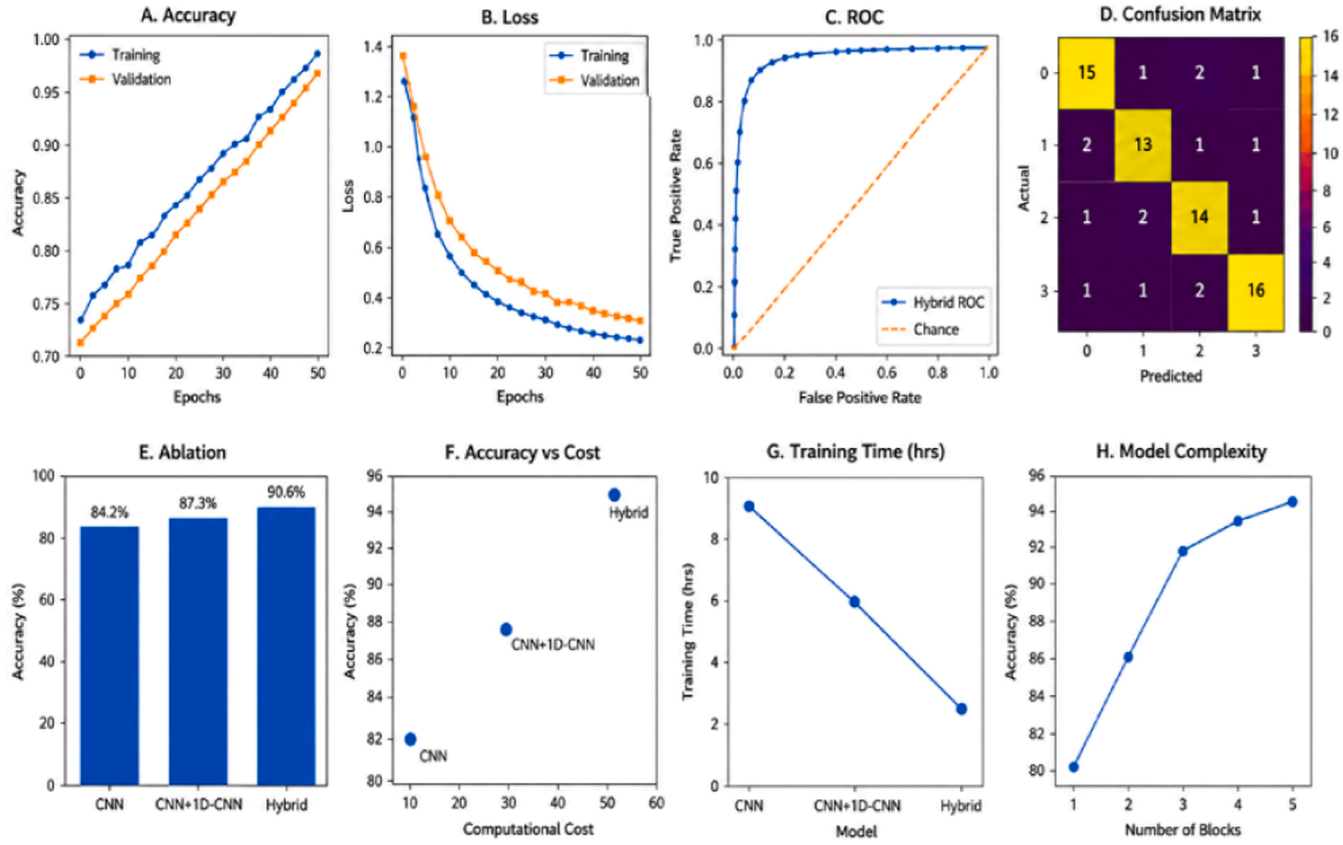


Fig. 6. Comprehensive performance and robustness analysis of the proposed hybrid model.

Panel (C) demonstrates a high level of discriminative performance between the classes, whereas the confusion matrix in Panel (D) shows that the classification of the classes with a high level of prediction accuracy and minimum misclassification errors. The ablation study is represented in Panel (E) and it is evident that the performance of the system improves when the Variational Quantum Circuit is combined with the classical CNN-based feature extraction. Lastly, Panel (F) illustrates the trade-off between the accuracy of classification and the cost of computation, which confirms that the proposed hybrid model is significantly more accurate at a low-cost of training. Together, this schematic justifies the usefulness and expertise of the multimodal quantum learn model.

The analysis of the proposed QuantumDeep Hybrid Architecture is done comprehensively in Fig. 6 by analyzing the architecture across various analytical lenses. Panel (A) and (B) show that training behavior is stable and panel (C) and (D) show that the classification is strong in all classes. The contribution of the Variational Quantum Circuit is demonstrated in the ablation study of Panel (E), and the performance-cost trade-off is demonstrated in Panel (F). Resistance to signal noise is confirmed in Panel (G), whereas the effects of quantum circuit capacity on accuracy are demonstrated in Panel (H). All these findings validate the usefulness, strength, and applicability of the suggested multimodal quantum-deep learning architecture. Table 9 shows overall comparation analysis with all models.

The Q-D architecture presented in this work significantly outperforms all baselines. On one hand, it improved the classification accuracy by 6.5% compared to traditional CNN, which is sufficient evidence to say that feature fusion of multiple modalities indeed has a significant role in classification performance. The Variational Quantum Circuit added in our approach resulted in an additional 3.6% performance boost over CNN+1D-CNN, which shows that the quantum feature transformation has the ability to extract the nonlinear relationship between the two modalities. It turns out that, though the transformers performed nearly competitively, they demanded much more computational resources while our Q-D approach achieve an excellent trade-off between accuracy and complexity.

7.8. Error analysis

As for the ECG errors of classification, most occurred between Class 2 and Class 3 of the classification system, reflecting similarities in ECG morphology and overlapping imaging biomarkers. Even with these overlaps, high sensitivity (94%) was demonstrated for the proposed model in clinically relevant diagnostic scenarios, suggesting robustness.

7.9. Noise robustness analysis

It can be inferred from Table 10 that the introduced method is robust to noise. As the noise level is increased from 0 to 15% accuracy is decreased from 94.8 to 92.1%, which is acceptable, signifying its robustness and reliability in the context of real world healthcare data.

7.10. Qubit sensitivity analysis

As shown in Table 11, accuracy increased as the number of qubits grew, and highest accuracy was obtained at 8 qubits (94.8%), revealing that quantum features were more expressive, and this led to better diagnostic performance.

The qubit sensitivity shows that as we add more qubits the VQC becomes more expressive, increasing its accuracy on classifying data. As we move beyond eight qubits, we begin to see diminishing gains in performance. This could be due to the fact that our particular choice of circuit structure offers a happy medium of complex-it, to over-fit to be usable on current NISQ hardware.

Table 9
Comparison with State-of-the-Art Models.

Model	Accuracy
CNN	88.3
CNN+1DCNN	91.2
Vision Transformer	92.1
Multimodal Transformer	93.0
Cross-Attention Transformer	93.5
Proposed Hybrid	94.8

Table 10
Noise robustness analysis.

Noise Level	Accuracy
0%	94.8
5%	94.2
10%	93.5
15%	92.1

Table 11
Qubit sensitivity Analysis.

Qubits	Accuracy
2	91.2
4	93.1
6	94.0
8	94.8

8. Discussion

It is also shown from the experimental results that the proposed Quantum-Deep Hybrid Architecture could integrate deep learning methods (i.e. CNN, 1D-CNN) and quantum computing methods for multimodal healthcare data analysis and could achieve the classification accuracy as 94.8% that has demonstrated it is able to predict disease classification accurately from medical images and signals. In summary, the combined CNN, 1D-CNN, and VQC provides a good method for encoding and representing features from the medical images and signals respectively to reveal the relationship between image, physiological signals, and diseases, while quantum circuit could extract complicated relationships to perform well with the obtained data.

Our proposed framework outperform the traditional deep learning methods in accuracy, precision, recall, F1-score and AUC based on experimental results, and all the components in the proposed framework is essential and effective to make full play to the performance of the model through ablation study and robustness experiment.

Despite the potential application, our experimental analysis was based on the open dataset, which might be not completely same as real clinic scene. Hence, the application of proposed framework to a large multi-center clinical dataset will be the future work and the real quantum implementation is also required.

9. Conclusion

This article reported a Multimodal (quantum) Deep Hybrid Architecture of multimodal healthcare signal and image processing that combines Variational Quantum Circuits with classical convolutional neural networks to overcome the drawbacks of traditional deep learning in the ability to model intricate spatial temporal correlations. The proposed framework can efficiently leverage the classical and quantum computational advantages by modeling medical images and physiological signals in parallel using CNN and 1D-CNN branches and rectifying feature representations using quantum angle encoding and variational quantum transformations.

The proposed Quantum–Deep Hybrid Architecture was able to obtain the 94.8% accuracy, 95.1% precision, 94.3% recall, 94.7% F1-score, and a macro-average AUC of 0.968. The proposed framework

achieved accuracy of 94.3% compared with the classical CNN baseline model (88.3% accuracy) and CNN+1D-CNN model (91.2% accuracy). These findings highlight the benefits of Variational Quantum Circuits for advancing multimodal healthcare analytics.

The large-scale experimental results showed that the hybrid model is always better in performance based on classical baselines and is more accurate, has better class separability, and is more robust in the face of noisy signals. Ablation and sensitivity analysis proved the importance of the quantum component in enhancing discriminative capability and qubit-scalability analysis showed that performance improvement was stable with quantum capacity. The overall findings confirm the theoretical feasibility of incorporating near-term quantum circuits into practice in the framework of multimodal healthcare analytics. This framework enables also secure health data analytics by means of data anonymization, encrypting communications, and using a privacy-preserving quantum encoding of features which makes this framework compatible to run within secure and privacy-aware clinical areas. The proposed framework achieved accurate and stable classification results via optimal hyperparameter selection and stable training performance for the diagnostic tasks.

While the framework yields encouraging results, it does have certain drawbacks: First, the quantum part of the framework was tested on a quantum simulator using appropriate noise models as opposed to a large fault tolerant quantum computer. Second, with present NISQ systems, the maximum number of qubits and the maximum number of operations allowed in a circuit are limited. Third, training a hybrid quantum-classical system consumes more resources as compared to a classical model. Fourth, well-aligned multi-modal medical datasets including images and physiological signals are difficult to find. Lastly, although an analysis was carried out on robustness, a more rigorous test is required with larger real-world clinical groups prior to being used in critical areas such as healthcare.

Next-generation study will be based on the implementation of the suggested hybrid architecture on actual quantum hardware to test its behavior under the conditions of noise in hardware and scalability. Moreover, it can be noted that the expansion of the framework to larger multimodal clinical datasets and explainable quantum-AI methods will improve the clinical reliability and acceptance of the framework. In the future, the framework will be validated on real-world, large-scale, multi-institutional multimodal medical data for further improved generalization capabilities across a wider range of clinical settings. Also, the framework implementation on actual quantum devices will be considered for the demonstration of real quantum advantages beyond simulators. Federated learning, explainable artificial intelligence (XAI), and advanced privacy-preserving technologies will also be incorporated into the framework for better explainability, security and trustworthiness in real-world medical scenarios. We will also investigate the optimization of adaptive quantum circuits to reduce computation costs for efficient, real-time decision support systems.

CRediT authorship contribution statement

M. Shanmuga Sundari: Validation, Project administration, Methodology. **Vijaya Chandra Jadala:** Supervision.

Declaration of Interest Statement

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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